ZZ-R-00690a (DSA-DM)
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INTERIM REVISION OF
FED. SPEC. ZZ-R-690
20 August 1958

INTERIM FEDERAL SPECIFICATION

RUBBER DAM, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The Government Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification ZZ-R-690.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rubber dams suitable for use in establishing a dry field in dental procedures.

1.2 Classification. Rubber dam shall be of the following types and sizes as specified (see 6.1).

Type I. Precut squares.

Size 1. Medium thickness.
Size 2. Heavy thickness.

Type II. Rolls

Size 1. Special heavy thickness.

FSC 6520

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, (For Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-601	Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Dam shall be made from rubber containing no less than 95 percent by volume of best quality new rubber having the physical requirements and conforming to the tests specified herein.

3.1.1 Color. Rubber dam shall be of a dark color so as to offer contrasting color to the tooth structures.

3.2 Style, design, and dimensions. Type I rubber dam shall be precut into squares. A slight curvature or temporary fusion of the cut edges to one another due to rotary cutting is permitted, provided separation can be accomplished easily by the operator without tearing of edges. Type II rubber dam shall be provided in roll form. Both types I and II rubber dams shall be suitable for use in mouths of dental patients to establish a dry field during dental procedures. Dimensions of types I and II rubber dams shall be as specified in table I.

Table I. Dimensions

	Length	Width	Thickness
Type I.			
Size 1.	6 ± 1/8 inches	6 ± 1/8 inches	0.007 to 0.009 inch
Size 2.	6 ± 1/8 inches	6 ± 1/8 inches	0.009 to 0.0115 inch
Type II.	18 feet minimum	6 ± 1/8 inches	0.0135 to 0.0155 inch

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3.3 Physical requirements.

3.3.1 Tensile strength. Rubber shall have a tensile strength of 4000 pounds minimum per square inch, when tested as specified in 4.3.

3.3.2 Ultimate elongation. Ultimate elongation shall be a minimum of 800 percent, when tested as specified in 4.3.

3.3.3 Resistance to accelerated aging. Tensile strength and ultimate elongation shall be no less than 50 percent of the original value, when tested as specified in 4.3.

3.3.4 Sterilization. The tensile strength and ultimate elongation shall be no less than 50 percent of the original value when tested as specified in 4.3, except that the time shall be increased to 2 hours, and a single sterilization cycle shall be performed.

3.4 Workmanship. Workmanship shall be first class throughout. Rubber dam shall have a smooth surface finish on both sides and shall be free from ingrained particles, blisters, and other imperfections which detract from its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component, or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and III (see 4.2.1 and 4.2.2) and table IV (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination purposes shall be one package of rubber dams (type I) or one roll (type II).

Table II. Examination Criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	4.0

4.2.2 Examination. The rubber dam shall be examined for defects including, but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Major:	
101	Type and size not as specified.
102	Color not dark.
103	Thickness not uniform.
104	Surface not free from ingrained particles, holes, blisters, or other imperfections.
Minor	
201	Precut individual rubber dams (type I) do not separate easily when taken from package.
202	Packaging and packing not as specified.
203	Finish not smooth.

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4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 inspection level S-1 except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one precut rubber dam (type I) or one roll of rubber dam (type II), and a minimum of three samples shall be taken.

4.3 Test methods. Tests shall be conducted in accordance with FED-STD-601, for the applicable test method as specified in table IV.

Table IV. Test Methods

Characteristic	Requirement	Test Method in FED-STD-601
Length	3.2	2411
Width	3.2	2121
Thickness	3.2	2011
Tensile strength	3.3.1	4111
Ultimate elongation	3.3.2	4121
Resistance to accelerated aging	3.3.3	7221
Sterilization	3.3.4	7411

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing thirty-six precut rubber dam squares (type I) or one roll of rubber dam (type II), of the dimensions specified in table I, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	48 units	192 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Marking shall include the legend:
"STORE IN A COOL PLACE (50°-80° F.)".

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend:
"STORE IN A COOL PLACE (50°-80° F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
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Preparing activity:
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Review information is current as of the date of this document. For future coordination of changes of this document, draft circulation should be based on the information in the current DODISS.

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